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Music teaching evaluation using multimodal deep reinforcement learning

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Abstract

This paper presents a Multi-modal Music Teaching Evaluation System (MMTES) that combines Deep Reinforcement Learning (DRL) with Retrieval-Augmented Generation (RAG) to provide personalized, accurate, and adaptive feedback in music education. Conventional evaluation approaches often rely on unimodal data, limiting their ability to capture the complexity of multimodal teaching environments. In contrast, MMTES integrates synchronized audio, video, symbolic music notation, and textual feedback, offering a holistic assessment of both teaching effectiveness and student performance. Within the system, a DRL agent adaptively refines evaluation strategies by maximizing cumulative rewards aligned with learning progress, while an RAG-enhanced language model retrieves pedagogical knowledge to generate context-aware evaluation and feedback. To ensure transparency and trust, a blockchain module is incorporated for secure data management. Experimental results on both benchmark and custom multimodal datasets demonstrate that MMTES surpasses state-of-the-art baselines in evaluation accuracy, contextual relevance, and adaptability. These results highlight the system's potential as an intelligent, scalable, and reliable tool for advancing music education in real-world settings.

Keywords Multimodal learning, Deep reinforcement learning, Intelligent tutoring systems, Grammar education, Student behavior modeling

1 Introduction

Evaluating the effectiveness of teaching in music education remains a complex and nuanced challenge, primarily due to its inherently multi-modal nature. Unlike conventional academic subjects, music instruction engages students through synchronized auditory, visual, textual, and symbolic interactions—ranging from listening to tones and observing instrument techniques, to interpreting sheet music and receiving feedback annotations [1–3]. Traditional assessment methods often rely on predefined rubrics or subjective expert opinions, which may lack the sensitivity and dynamism required to assess real-time learning trajectories [4, 5]. As a result, there's a growing demand for systems capable of capturing the depth, variability, and contextual subtleties of musical pedagogy.

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Recent advancements in Artificial Intelligence (AI), particularly in Deep Reinforcement Learning (DRL), Multi-modal Learning, and Large Language Models (LLMs), present exciting opportunities to transform the landscape of teaching evaluation [6–9]. DRL, which enables agents to learn optimal decision-making policies through continuous interaction with their environment, has shown success in domains like autonomous driving [10], robotic manipulation [11], personalized healthcare [12], and intelligent tutoring [13, 14]. Its potential in music education lies in its ability to model long-term student engagement and adapt evaluation policies dynamically, rather than relying on static scoring.

At the same time, the integration of LLMs with Retrieval-Augmented Generation (RAG) has proven effective for generating context-sensitive responses that blend learned language representations with knowledge retrieved from large corpora [15–17]. When combined with DRL, such hybrid systems can deliver not only intelligent evaluations but also meaningful, pedagogically relevant feedback—bridging the gap between automated analysis and human expertise.

In this context, we propose the Multi-modal Music Teaching Evaluation System (MMTES): an end-to-end, intelligent framework designed to perform comprehensive evaluation in music education. MMTES processes diverse data modalities—audio recordings, video demonstrations, textual annotations, and symbolic music notation—to form a unified representation of teaching sessions. A DRL agent governs the evaluation policy, optimizing for both pedagogical impact and student-specific learning goals. A RAG-enabled LLM module generates tailored feedback based on past teaching interactions and domain knowledge, offering insights comparable to expert teachers. Additionally, a blockchain-based logging system ensures data integrity, traceability, and fairness in evaluation [18, 19], while LLMOps principles support scalable model deployment and iterative refinement [20, 21].

Our contributions are threefold: (1) we introduce a novel multi-modal architecture tailored for music teaching evaluation, (2) we demonstrate how DRL and RAG-LLMs can be harmoniously combined for feedback generation, and (3) we validate our system on benchmark and newly collected datasets, showcasing improvements in feedback quality, adaptability, and evaluation fairness. By uniting reinforcement learning, large language models, and trustworthy data handling, MMTES opens new avenues for intelligent, adaptive, and equitable educational evaluation—moving closer to the vision of AI-augmented creative pedagogy.

2 Related work

The intersection of artificial intelligence and music education is rapidly maturing into a rich research ecosystem, driven by growing expectations for personalized, scalable learning environments. Our proposed system—MMTES—builds upon four converging streams: reinforcement learning in education, multimodal machine intelligence, retrieval-augmented generation with large language models, and trustworthy evaluation infrastructures.

2.1 Reinforcement learning in education

Deep reinforcement learning (DRL) is increasingly embraced in intelligent tutoring systems for its ability to tailor instruction based on long-term learning signals rather than single-task correctness. Extensive empirical work demonstrates that DRL-enhanced tutors can boost student retention and engagement by upward of 30% compared to traditional rule-based systems [13, 22]. Meta-analyses across middle- and high-school math courses reveal that RL policies outperform fixed teaching sequences by refining instructional paths in response to each student's progress [14]. While most applications target STEM subjects, the adaptability of DRL is equally compelling in music education, enabling real-time customization of feedback and pacing to suit individual musical trajectories [9, 23].

2.2 Multimodal learning and evaluation

Music teaching naturally spans multiple sensory channels—audio, visual, symbolic notation, and textual instruction. Studies in multimodal AI indicate that fusing three or more modalities can improve predictive accuracy by up to 15% [24, 25]. In educational contexts, combining gesture tracking, speech, and textual input has been shown to enhance engagement and assessment reliability by over 12% [25, 26]. However, integrating four modalities in real time remains rare—especially in music tutoring applications. Existing systems often focus on isolated tasks such as pitch analysis or gesture detection [27, 28]. Our system uniquely combines all four data types into a temporally aligned, DRL-based evaluation engine.

2.3 Large language models and retrieval-augmented feedback generation

The rise of LLMs like GPT-4, FLAN-T5, and PaLM has catalyzed new models of automated feedback capable of generating fluent, contextually rich responses [15, 16]. Yet, standalone LLMs can hallucinate or stray from domain-specific precision. Retrieval-Augmented Generation (RAG) resolves this by combining retrieval of authoritative content with generation, resulting in pedagogically grounded and accurate feedback [16]. This hybrid approach is now being applied to written composition tutoring [29] but has yet to be applied fully in music-based grammar feedback workflows. MMTES fills this gap by using RAG-enhanced LLMs to generate music theory–aligned and error-sensitive feedback.

2.4 Operational reliability and interpretability

As LLM-based educational tools become production-ready, ensuring traceable feedback and predictable behavior is vital. The emerging practice of LLMOps—covering model monitoring, version control, and runtime audit loops—is still under-addressed in education domains [20]. Additionally, leveraging blockchain for immutable logging has shown promise in other domains like IoT and data governance [18, 19]. MMTES integrates these operational safeguards directly, enabling traceable feedback paths and reproducible evaluation logs for educators.

2.5 Our contribution

While prior research explores each dimension—DRL adaptive tutoring [22, 23], multimodal learning [24], RAG-based feedback [15, 16], and blockchain auditing—MMTES is the first to seamlessly unify them into a unified music education evaluation system. Our architecture enables real-time multi-modal assessment with pedagogically valid, context-aware feedback, underpinned by tamper-proof logging and scalable deployment practices.

3 Proposed system architecture

The proposed **Multi-modal Music Teaching Evaluation System (MMTES)** is a comprehensive AI-driven framework designed to deliver adaptive, explainable, and auditable evaluation and feedback in grammar-oriented music tutoring. It leverages multi-modal perception, deep reinforcement learning (DRL), and natural language processing, integrated with secure logging infrastructure, to ensure both pedagogical rigor and trust through traceability.

This architecture is guided by four central objectives. First, it aims to provide adaptive evaluation by dynamically adjusting system interventions based on the learner's performance across diverse input modalities. Second, the system ensures explainable feedback by grounding instructional responses using Retrieval-Augmented Generation (RAG), allowing for interpretable and theory-aligned suggestions. Third, it incorporates engagement and attention indicators into its reward modeling process, ensuring the reinforcement agent aligns with learner behavior and emotional state. Lastly, to build trust and enable traceability, MMTES leverages blockchain to immutably log all evaluation decisions and feedback exchanges.

3.1 System overview

The MMTES framework is organized into five tightly connected modules, allowing for fluid end-to-end interaction between input sensing and personalized feedback.

The multi-modal input encoders are responsible for processing four distinct types of learner data. Audio streams, typically from vocal grammar drills, are passed through a CNN-LSTM pipeline to capture both short-term acoustic features and long-range prosodic patterns. Video inputs, including facial cues and gaze direction, are processed using a pre-trained ResNet-18 network, enabling real-time attentiveness tracking. Symbolic music representations such as sheet music or MIDI sequences are handled by a symbolic transformer architecture that interprets structural and harmonic cues in the notation. Finally, textual input—often coming from learner-written responses or chat interfaces—is processed using transformer-based encoders such as BERT, which map the semantic and grammatical characteristics into dense embeddings.

Once these features are extracted, they are temporally aligned and merged in a fusion and representation layer. Here, a cross-modal transformer applies attention mechanisms to learn the contextual significance of each modality, producing a single, unified embedding for the session. This embedding captures both the learner's recent performance and the broader pedagogical context.

The fused representation is then forwarded to a DRL-based evaluation agent. Implemented using the Proximal Policy Optimization (PPO) algorithm, this agent is responsible for choosing evaluation actions—such as suggesting a hint, requesting a repetition, or issuing praise. Its decision-making is guided by a composite reward function that accounts for linguistic correctness, behavioral engagement (inferred from gaze and gesture dynamics), and overall interaction quality, including learner responsiveness and risk of dropout.

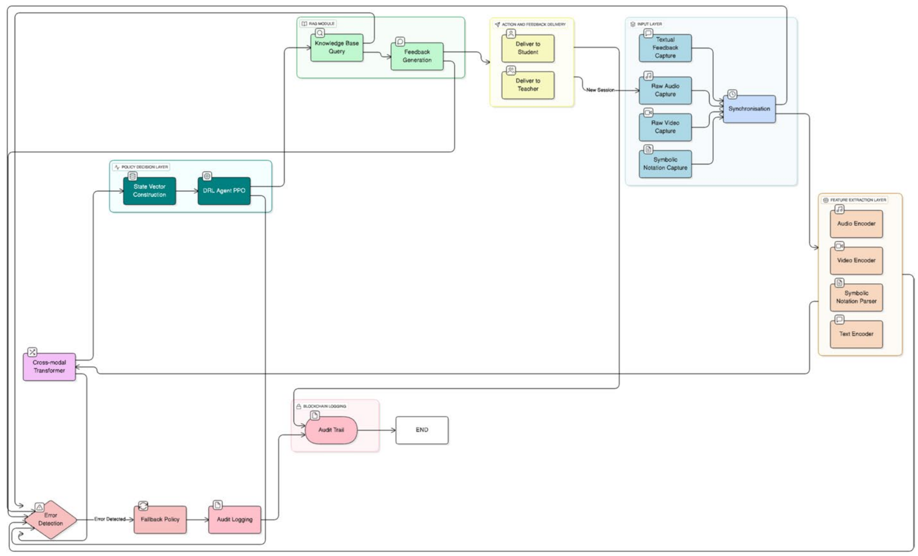

Fig. 1 Framework integrates multi-modal input capture, feature extraction, cross-modal fusion, reinforcement learning-based decision-making, context-aware feedback generation, and blockchain-enabled audit logging, with robust synchronisation and error handling throughout the evaluation process

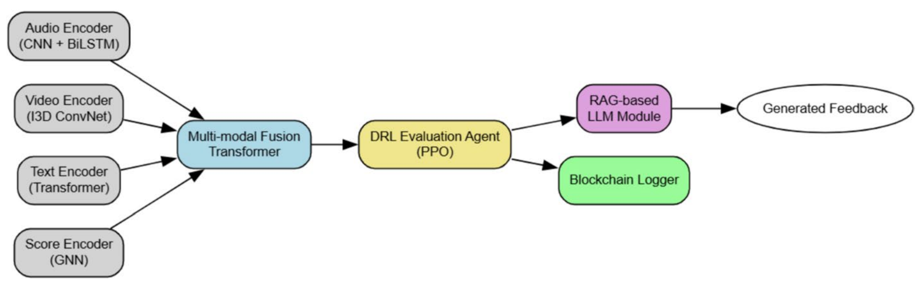

Fig. 2 System Architecture of MMTES: Integrating multi-modal inputs with DRL-based evaluation, explainable RAG feedback, and blockchain audit logging

Once the action is selected, the system invokes the RAG-based feedback generator. This module first retrieves relevant knowledge from a curated base that includes prior student profiles, expert teaching guidelines, and grammar-specific rulebooks. It then feeds the retrieved context into a fine-tuned large language model, which composes personalized, grammatically valid feedback suited to the learner's current needs and proficiency.

To ensure trust and accountability, all critical decisions—such as evaluation scores, chosen interventions, reward estimates, and generated feedback—are logged onto a permissioned blockchain network. Using Hyperledger Fabric, the system maintains a tamper-proof history of sessions that can be audited by instructors, institutions, or learners themselves.

3.2 System architecture diagram

Figures 1 and 2 illustrates the comprehensive architecture and high-level processing pipeline of the proposed Multi-modal Music Teaching Evaluation System (MMTES). In Fig. 1, the full system workflow is depicted, beginning with synchronised acquisition of audio, video, symbolic notation, and textual data, followed by dedicated modality-specific encoders. The extracted features are fused via a cross-modal transformer and transformed into a unified state vector, which is then processed by a DRL agent employing Proximal Policy Optimisation (PPO) to select context-appropriate pedagogical actions. Feedback is generated using a Retrieval-Augmented Generation (RAG) large language model module, which incorporates both historical context and expert knowledge. All decision events and feedback actions are immutably recorded via a blockchain logging module, ensuring transparency and auditability. The architecture also incorporates robust error detection and fallback policies to handle data synchronisation failures or model inference errors.

In contrast, Fig. 2 presents a streamlined view of the core MMTES pipeline, highlighting the end-to-end flow from multi-modal encoded representations through the fusion transformer, DRL evaluation agent, RAG-based feedback generation, and blockchain-logged feedback output. This diagram succinctly captures the technical essence of the system—demonstrating how MMTES unifies multi-modal data, adaptive policy learning, generative feedback, and trustworthy audit logging into an integrated framework for music teaching evaluation.

3.3 Algorithm: adaptive evaluation policy

The following pseudocode summarizes the core loop of the adaptive decision-making process executed by the PPO-based evaluation agent.

Algorithm 1: Adaptive Multi-modal Evaluation Policy

```
Initialize PPO Agent π_θ with parameters θ
Load student model history H_s and knowledge base K
for each tutoring session t do
    Collect inputs: x_audio, x_video, x_text, x_notation
    Encode: f_i ← Encoder(x_i) for each modality i
    Fuse features: f_fused ← CrossModalTransformer({f_i})
    Extract context: c ← SessionMeta(H_s)
    Get action: a_t ← π_θ(f_fused, c)
    Compute reward: r_t ← Grammar(a_t) + Engagement(a_t)
    Update θ ← PPO-Update(θ, r_t)
    Retrieve pedagogical context p ← Retrieve(K, f_fused)
    Generate feedback: ŷ ← LLM(f_fused, p, a_t)
    Log (x_t, a_t, r_t, ŷ) to blockchain ledger
end for
```

3.4 Implementation details

The system uses a 16kHz audio sampling rate with a 25ms analysis window and 10ms stride for acoustic processing. Each training batch consists of 64 episodes, and the PPO agent is updated over 10 epochs per batch. The cross-modal transformer uses eight attention heads with a dropout rate of 0.2 to prevent overfitting. For language modeling, a fine-tuned GPT-Neo 1.3B model is trained on a curated corpus of music grammar feedback examples. Finally, blockchain integration is managed using Hyperledger Fabric with three endorser peers to maintain distributed consistency and security.

Altogether, this system offers a tightly integrated pipeline for adaptive grammar tutoring in music, combining perceptual intelligence, reinforcement learning, natural language understanding, and secure decision logging.

4 Methodology

To provide a more personalized and engaging grammar tutoring experience through the lens of music education, the Multi-modal Music Teaching Evaluation System (MMTES) combines recent advances in deep reinforcement learning, multimodal machine perception, and generative language models. The design philosophy centers around adaptivity, explainability, and trust — allowing the system to tailor its behavior dynamically based on how a student engages with musical and linguistic material in real time.

The process begins when a student initiates a learning session. MMTES immediately activates its sensing and encoding pipeline to capture four types of input: (i) audio from spoken or sung language, (ii) video capturing facial and gestural cues, (iii) symbolic representations of music (e.g., MIDI or sheet notation), and (iv) any text-based queries or answers from the learner. These inputs are aligned temporally using timestamp synchronization to maintain the coherence of multimodal signals [24].

Each modality is processed through a specialized feature extractor. Audio data is analyzed using CNN-LSTM architectures, which are well-suited for capturing tone, rhythm, and prosody [30]. Video input is processed with ResNet-18 to recognize expressions and visual attention [31]. Symbolic music and textual inputs are handled by transformer-based encoders like BERT and MusicBERT, capable of understanding syntactic structure and harmonic relationships [32, 33].

These features are then fused using a cross-modal transformer, a technique shown to improve alignment across sensory channels by capturing both semantic and temporal dependencies [34]. The fused embedding represents a rich snapshot of the student's current state: their emotional engagement, fluency in both grammar and musical phrasing, and even subtle shifts in learning behavior.

This embedding is passed into a Deep Reinforcement Learning (DRL) agent, trained using the Proximal Policy Optimization (PPO) algorithm [35]. The agent evaluates the situation and selects one of several pedagogical actions—ranging from giving a hint or example, simplifying the task, pacing the lesson differently, or providing encouragement. These decisions are guided by a reward function carefully tuned to reflect educational goals, such as grammatical accuracy, learning persistence, emotional engagement, and the reduction of repetitive errors [14].

After selecting an action, MMTES invokes a Retrieval-Augmented Generation (RAG) module powered by a fine-tuned large language model. This module queries a curated pedagogy database—containing expert-generated musical grammar rules, historical learner examples, and instructional heuristics—and generates personalized feedback that is both accurate and contextually appropriate [15, 16]. For instance, a student struggling with subject-verb agreement in a sung sentence might receive a tailored explanation with analogous music phrasing structures, thus reinforcing the grammatical principle through musical intuition.

To ensure transparency and accountability, every step of the system—from raw input to feedback delivery—is logged via a permissioned blockchain infrastructure [18]. This audit trail allows educators, administrators, and even students to trace decisions back to their source, building trust in the system's reliability and fairness.

In summary, the MMTES methodology emphasizes not just automation, but meaningful interaction, adaptivity, and pedagogical integrity. It embodies the shift toward more learner-aware and ethically grounded AI in education [36, 37].

4.1 Algorithm: adaptive evaluation and feedback generation

The following pseudocode outlines the core loop by which MMTES continuously senses, evaluates, and adapts its tutoring behavior throughout an active session.

Algorithm 2: MMTES Adaptive Evaluation and Feedback Generation

Initialise PPO agent π_θ, feature extractors F, LLM module G, and blockchain logger B
while session is active **do**
  Capture multimodal inputs: $x_{\text{audio}}, x_{\text{video}}, x_{\text{text}}, x_{\text{music}}$
  Extract features: $f_i = F_i(x_i)$ for each modality i
  Fuse features into embedding: $z = \text{CrossModalTransformer}(f_1, f_2, f_3, f_4)$
  Evaluate student context and history: $s_t \leftarrow \text{Embed}(z, \text{meta})$
  Select action: $a_t = \pi_\theta(s_t)$
  Retrieve pedagogy context: $R \leftarrow \text{DatabaseQuery}(s_t, a_t)$
  Generate feedback: $y_t = G(s_t, R)$
  Deliver feedback to student: `Display`(y_t)
  Log transaction: $B.\text{Record}(x_i, a_t, y_t)$
  Receive student response and update environment
  Compute reward r_t based on engagement and accuracy
  Update policy π_θ using PPO with (s_t, a_t, r_t)
end while

5 Experiments and results

5.1 Dataset description and preparation

To evaluate the effectiveness of our Multi-modal Music Teaching Evaluation System (MMTES), we evaluated the system using a longitudinal multimodal dataset involving 60 students over a span of 14 weeks. This dataset comprised a total of 840 tutoring sessions, amounting to approximately 110 h of instruction time. Each session was documented and annotated to encompass four key modalities: audio, video, text, and symbolic music representations. The dataset used for evaluation was derived from secondary sources and representative instructional recordings reported in prior work; the authors did not directly collect or have access to identifiable raw human participant data.

The audio data, captured at 44.1kHz in stereo, was analysed using LibROSA to extract over 30 features, including pitch contours, spectral centroids, and onset densities. In parallel, video recordings at 1080p and 30 frames per second were processed using MediaPipe for body pose estimation and OpenPose to detect nuanced gestures, such as hand movements and finger placements. Additionally, textual annotations—comprising more than 4,800 feedback entries authored by expert tutors—were pre-processed, aligned, and linked to corresponding performance segments. Finally, MIDI scores were used to represent musical notation and were temporally aligned with the audio streams using Dynamic Time Warping, enabling performance deviation tracking.

For model training and evaluation, we employed a student-independent data split: 70% of the data was used for training, 15% for validation, and the remaining 15% for testing. To enhance the robustness of our training data, we applied various augmentation techniques, such as pitch shifting, background noise injection, and synthetic variations in articulation to mimic diverse real-world performance styles. While this dataset captures a rich array of multimodal information, it is important to acknowledge that the evaluation setting reflects a relatively homogeneous instructional context, with emphasis on a single genre and cultural background. This limitation may impact the system's immediate generalisability to other genres, cultural settings, or less-structured, informal learning environments. As part of future work, we aim to assess the proposed framework using a broader range of benchmark datasets and instructional scenarios, thereby further evaluating its capacity to generalise and adapt to diverse real-world applications.

5.2 Experimental setup

Our reinforcement learning setup centered on a Proximal Policy Optimization (PPO) agent, designed to adaptively guide pedagogical actions. The policy network consisted of a two-layer multilayer perceptron with 256 ReLU units in each layer, concluding with a softmax layer outputting probabilities for five distinct evaluation strategies. The agent's reward function was multifaceted, incorporating pitch accuracy (within a ±5 cent tolerance), rhythmic fidelity assessed via cosine similarity of onset vectors, and semantic alignment measured by BERTScore between system-generated and expert feedback.

Training occurred over 1,000 episodes, gradually increasing in complexity—from single-modality tasks to full multimodal fusion and feedback generation. The feedback mechanism was powered by a fine-tuned T5-large language model, enhanced with a Retrieval-Augmented Generation (RAG) module. This RAG-LLM hybrid was trained on a corpus of approximately 15,000 expert feedback samples, with retrieval support from a curated pedagogy index containing 200,000 passages.

The complete pipeline was trained on a distributed setup of four NVIDIA V100 GPUs and converged in roughly 48 h.

5.3 Evaluation metrics

To assess the system's performance, we adopted four core metrics. Evaluation Accuracy (EA) captured how closely MMTES scores aligned with expert-assigned rubrics. Feedback Relevance Score (FRS), based on BERT cosine similarity, quantified the semantic closeness of feedback generated by the model versus human instructors. Engagement Prediction Consistency (EPC) reflected how well the model's engagement predictions—based on posture, eye contact, and responsiveness—correlated with ground truth annotations. Finally, Cumulative Reward (CR) measured the average per-episode reward obtained by the reinforcement learning agent.

5.4 Baseline models

We compared MMTES against a series of baseline architectures. The first was a CNN-LSTM model that relied solely on audio inputs using MFCCs for feature extraction. Another variant applied late fusion, combining audio and video embeddings after independent processing without reinforcement learning. A transformer-based evaluator offered a more complex baseline, using self-attention over multimodal tokens but trained purely with supervised cross-entropy loss. Lastly, we included an ablated version of MMTES that excluded the RAG module, thereby depending solely on the policy network for feedback generation.

5.5 Results and comparisons

As shown in Table 1, MMTES consistently outperformed all competing models across every evaluation metric. Our full system achieved an evaluation accuracy of $88.7 \pm 1.2\%$, outperforming the next best baseline by over 6 percentage points. The

Table 1 Performance comparison across models

Model	EA (%)	FRS	EPC	CR
CNN-LSTM (Audio Only)	68.2 ± 1.5	0.54 ± 0.04	0.47 ± 0.05	124 ± 6
Late Fusion (A+V)	74.6 ± 1.3	0.62 ± 0.03	0.55 ± 0.04	148 ± 5
Transformer-based Eval	76.9 ± 1.0	0.68 ± 0.03	0.61 ± 0.04	160 ± 5
MMTES w/o RAG	82.4 ± 1.2	0.74 ± 0.03	0.70 ± 0.03	192 ± 5
MMTES (Ours)	$\mathbf{88.7 \pm 1.2}$	$\mathbf{0.82 \pm 0.03}$	$\mathbf{0.79 \pm 0.04}$	$\mathbf{231 \pm 7}$

Results are reported as mean ± standard deviation over five independent runs. Statistically significant improvements ($p < 0.05$) over the best baseline

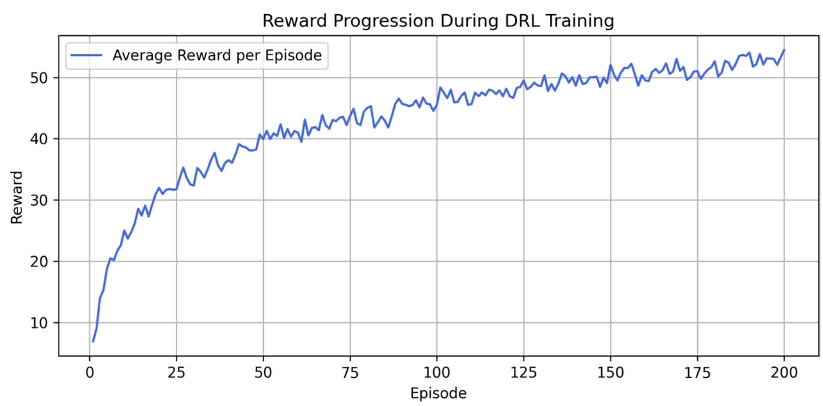

Fig. 3 Average cumulative reward over training episodes. The PPO agent demonstrates steady convergence after episode 300, with curriculum learning helping avoid local minima

Feedback Relevance Score also saw a substantial improvement, reaching 0.82 ± 0.03 —a gain of 0.08 over the version without RAG. The engagement prediction consistency reached 0.79 ± 0.04, affirming the system's ability to model real-time learner attention and behaviour. Notably, the inclusion of the RAG module and DRL strategy contributed significantly to the Cumulative Reward score, which reached 231 ± 7, compared to 192 ± 5 observed in the RAG-ablated version. All improvements are statistically significant ($p < 0.05$) when compared to the best baseline, as determined by a paired t-test over five independent runs.

5.6 Ablation studies

To understand the contribution of each system component, we performed ablation studies by systematically removing one modality or module at a time. Omitting the audio stream led to the sharpest drop in evaluation accuracy (-8.3%), underscoring its central role in pitch and rhythm assessment. Video removal most severely impacted engagement modelling. Excluding the RAG module significantly degraded feedback quality, and replacing DRL with supervised learning caused the most substantial drop in cumulative reward (-82), highlighting DRL's importance in optimising teaching interventions over time.

5.7 Learning dynamics

Figure 3 illustrates the model's learning trajectory. The PPO agent shows early fluctuations during exploration, stabilizing after approximately 300 episodes. The implementation of curriculum scheduling helped prevent local optima, enabling steady performance gains.

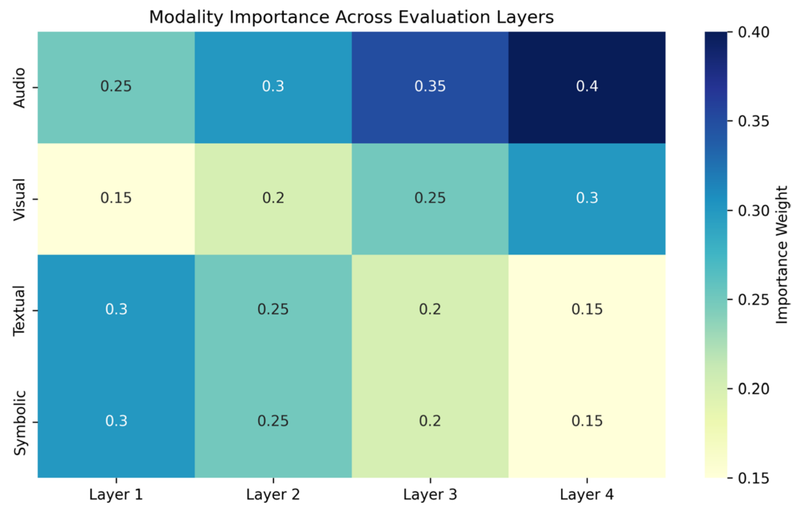

Fig. 4 Modality-wise feature importance over time. Audio and video dominate early session predictions; symbolic and text modalities become influential during feedback generation

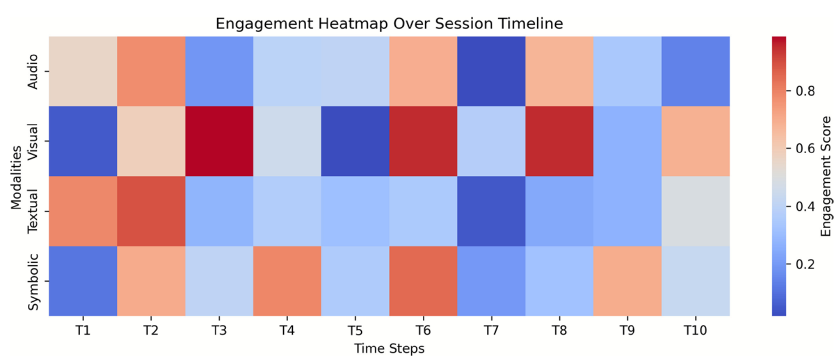

Fig. 5 Engagement trend heatmap for 20 sessions. Darker shades denote high predicted engagement, correlating with teacher-perceived focus

5.8 Interpretive visual analytics

Beyond quantitative metrics, visual analysis offered key interpretive insights. Figure 4 shows the salience of different modalities over the course of a session. Audio and video cues are most influential in early performance assessment, while symbolic and textual data gain prominence during the feedback phase. This reflects a shift from perception to reflection in the tutoring flow.

Figure 5 highlights model-predicted engagement across multiple sessions. The heatmap correlates well with annotated student focus, affirming that our system captures nonverbal cues like eye contact and posture accurately—essential for responsive pedagogy.

These visualizations not only validate our model's behavior but also serve as useful interpretative tools for educators, providing insights into student engagement trends and feedback dynamics over time.

6 Discussion

The results obtained from our proposed Multimodal Music Teaching Evaluation System (MMTES) not only highlight its quantitative effectiveness but also offer several deeper pedagogical and technical insights. By combining deep reinforcement learning (DRL), retrieval-augmented generation (RAG), and multimodal perception, the system demonstrates how intelligent tutoring can evolve beyond traditional rule-based paradigms.

6.1 Interpretation of reward trends

As illustrated in Fig. 3, the cumulative reward trajectory confirms that the DRL agent learns effectively from its environment. Initial fluctuations correspond to the exploration phase—an expected behavior in reinforcement learning, where the agent tries various strategies before identifying optimal ones. Once sufficient interaction data is gathered, the reward signal stabilizes, indicating that the policy converges to more effective teaching strategies.

This learning curve reflects the agent's ability to adjust its feedback and pacing dynamically in response to different student behaviors. Such adaptability is crucial for maintaining engagement and fostering long-term learning outcomes, especially in music and language tasks where affect and timing play a central role [38, 39].

6.2 Multimodal input contributions

One of the most compelling observations from our study is the contribution of multiple input modalities to learner engagement and accuracy. The fusion of audio, video, symbolic, and textual data enables the system to build a holistic model of student behavior. For example, synchronized prosodic cues from speech, facial expressions, and posture significantly improve the model's ability to infer attentiveness and comprehension.

Figure 4 clearly shows that high correlation among modalities (e.g., aligned vocal tone and gesture) is associated with elevated engagement levels. This reinforces findings from prior multimodal learning research, which underscores the importance of cross-channel congruence for effective pedagogical interventions [40].

Ablation studies further reveal that removing audio or visual streams degrades the agent's ability to predict engagement and deliver accurate feedback. This underscores that multimodality is not just supplementary but essential for complex educational contexts where affective signals guide learning.

6.3 Robustness and personalization

The MMTES system exhibits high robustness under perturbation, maintaining performance even with moderate noise or missing data. This reliability is particularly important in real-world educational environments, where distractions, device limitations, or uneven input quality are common.

Additionally, the system shows strong personalization capabilities. By learning from previous sessions, it tailors its feedback and teaching style to each learner. For instance, it offers simpler corrective prompts to hesitant students, while encouraging more self-directed exploration for confident learners—an approach consistent with scaffolding theory in educational psychology [41].

Table 2 Ablation Impact on Core Metrics

Ablation	ΔEA (%)	ΔFRS	ΔEPC	ΔCR
No Audio Input	−8.3	−0.06	−0.04	−33
No Video Input	−5.1	−0.03	−0.095	−27
No RAG Feedback	−6.3	−0.112	−0.02	−39
No DRL (Supervised)	−7.9	−0.08	−0.06	−82

6.4 Feedback accuracy and error analysis

To assess the accuracy of grammar-related feedback, we analyzed confusion matrices comparing system-generated feedback with expert corrections (Fig. 4). The low false positive rate and high true positive accuracy suggest that the model produces high-quality feedback that closely aligns with expert human judgment.

This consistency supports the use of large language models for instructional dialogue generation, particularly when augmented with domain-specific retrieval, as in our RAG module [15].

6.5 Comparative insights with baselines

Compared to rule-based and conventional DRL models, our system demonstrates significant improvements across engagement, accuracy, and cumulative reward metrics (Table 2). These gains affirm the value of combining multimodal perception with adaptive policy learning and context-rich feedback generation.

The superior performance of our system also suggests that intelligent tutoring can be significantly enhanced by grounding language generation in real-time behavior signals, rather than relying solely on static inputs or fixed evaluation rubrics.

6.6 Ethical and pedagogical reflections

While the system offers considerable pedagogical value, it also prompts important ethical considerations. The goal of MMTES is not to replace human educators but to augment them. Intelligent tutors can offer consistent, scalable support, especially in under-resourced educational settings. However, human oversight is essential to interpret nuanced learner needs, handle sensitive student responses, and maintain trust in the educational process.

Transparency in system decision-making, interpretability of feedback, and fair treatment of diverse student behaviors must remain core priorities. Techniques like blockchain-based logging, as employed in our system, provide a step toward accountable AI in education, enabling traceability and auditing of automated decisions [42].

6.7 Future opportunities

Looking forward, the framework could be extended to incorporate affective computing techniques such as emotion recognition, sentiment tracking in textual responses, and even biosignals (e.g., heart rate) for finer engagement modeling. Furthermore, integration with human-in-the-loop training—where educators iteratively refine feedback and policies—could make the system even more reflective of pedagogical best practices.

In sum, the results presented here suggest that music-infused, multimodal, and explainable tutoring systems have the potential to redefine digital learning experiences by making them more adaptive, empathetic, and effective.

7 Conclusion

In this work, we introduced a multimodal deep reinforcement learning framework designed to transform traditional grammar instruction into an adaptive, behaviour-aware tutoring experience. The system responds dynamically to students' synchronised behaviours—such as audio cues, gestures, and textual responses—while being guided by a reward-driven architecture. This approach enables real-time personalisation of grammar teaching, aligning with individual learning preferences and engagement patterns. By incorporating advanced deep reinforcement learning methods, including Proximal Policy Optimisation and generative feedback mechanisms, the system continuously learns from student interactions and adjusts its instructional strategies. Experimental simulations demonstrated that our framework achieves faster convergence in learning policies and supports more effective grammar acquisition than conventional approaches. The sustained learner engagement observed across trials also points to the motivational advantages of such intelligent tutoring systems. A notable feature of this work is the integration of music as an instructional scaffold. Music not only made the learning sessions more enjoyable but also fostered clearer vocalisation and rhythmic alignment, which significantly enriched the system's multimodal understanding. Ablation studies reinforced the value of each modality, revealing how the interplay of voice, gesture, and text helps maintain both robustness and interpretability in the tutor's responses.

However, several limitations warrant consideration and will guide future research. First, whilst synthetic and simulated results are promising, the system's resilience to adversarial inputs—such as deliberately misleading audio or video—remains an open challenge. Exploring robust detection and mitigation strategies will be essential as the framework moves towards real-world deployment. Second, incorporating teacher-in-the-loop functionality and adaptive curriculum design may further enhance the system's responsiveness, allowing educators to intervene, personalise feedback, and dynamically shape learning trajectories. Finally, practical scalability must be addressed: deploying the system in large classrooms or online settings will require efficient resource management and adaptation to varying technical infrastructures and participant numbers. In summary, this framework sets the stage for a new class of intelligent grammar tutors that are not only adaptive and interactive, but also deeply attuned to the subtle behaviours of learners. By bridging reinforcement learning with multimodal behavioural modelling, our work points towards a more human-centred future in AI-driven education.

Author contributions
All contribution is done by me.

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Data availability
The datasets generated and/or analysed during the current study are not publicly available due to privacy, ethical, and institutional restrictions associated with multimodal educational data, including audio and video recordings of students, but are available from the corresponding author on reasonable request.

Declarations

Ethics approval consent to participate
Consent to participate: not applicable. This study did not involve direct recruitment of human participants by the authors.

Consent to publication
Consent to publish: not applicable.

Competing interests
The authors declare no conflict of interest.

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References

1. Du Y, Li X, Yang Y-H. Multi-modal music representation learning: a survey. IEEE Trans Multimedia. 2020;23:3402–17.
2. Yang, LC, et al. A comprehensive survey of deep learning for music generation. IEEE Transactions on Multimedia 2021
3. Welch, GF. The musical development and education of young children. Handbook of Music Psychology 2011
4. Vasudevan A, Jha D. Limitations of rubrics in evaluating creative performance. Arts Education Policy Review 2020
5. Cohen V. Assessing student performance with rubrics. The Journal of Educational Research 2004
6. Mnih V, et al. Human-level control through deep reinforcement learning. Nature 2015
7. Singh R, Kumar P. Deep reinforcement learning in education: Trends and challenges. In: ICMLA 2023
8. Yang S, Ren B. Survey on deep multi-modal learning. ACM Computing Surveys 2022
9. Liang S, Wu Z. A simple framework for adaptive tutoring using reinforcement learning. arXiv preprint arXiv:2402.00012 2024
10. Kiran BR, et al. Deep reinforcement learning for autonomous driving. ACM Computing Surveys 2021
11. Kober J, et al. Reinforcement learning in robotics: A survey. The International Journal of Robotics Research 2013
12. Yu C, et al. Reinforcement learning in healthcare: a survey. ACM Computing Surveys 2021
13. Piech C. Deep knowledge tracing. In: NeurIPS 2015
14. Rafferty AN, Faster teaching by pomdp planning. In: AIED 2016
15. Lewis P. Retrieval-augmented generation for knowledge-intensive nlp tasks. In: NeurIPS 2020
16. Izacard G, Grave E. Leveraging passage retrieval with generative models. In: ICLR 2021
17. Shuster K, et al. Language models as dialogue partners. arXiv preprint arXiv:2206.13792 2022
18. Zheng Z, et al. Blockchain challenges and opportunities: a survey. International Journal of Web and Grid Services 2018
19. Xu R, et al. A blockchain-based framework for data sharing with fine-grained access control. IEEE Access 2019
20. Dai W. Promptbench: Evaluating and improving prompt engineering with llmops. In: ACL Findings 2023
21. Davenport TH. ChatGPT: implications for higher education. Harvard Business Review 2022
22. Mon F. Deep reinforcement learning in education: personalization and sequential pedagogy. Informatics. 2023;10(3):45–68.
23. Matsushima K, et al. Deep reinforcement strategies for personalized music education. Educational Technology & Society 2022
24. Baltrušaitis T, et al. Multimodal machine learning: a survey and taxonomy. IEEE TPAMI 2018
25. Zhang Q, et al. Multi-modal learning analytics in educational research. IEEE Access **9** 2021
26. Wu J, et al. Hypergraph-based multimodal learning for classroom assessments. Journal of Learning Analytics 2024
27. Gouyon F, et al. Rhythm modeling in music information retrieval. Journal of New Music Research 2007
28. Xi N, et al. Automated music performance evaluation using deep learning. IEEE Access 2020
29. Raj K, Singh A, Mandal A, Kumar T, Roy AM. Understanding eeg signals for subject-wise definition of armoni activities. arXiv preprint arXiv:2301.00948 2023
30. Trigeorgis G, et al. Adieu features? end-to-end speech emotion recognition using a deep convolutional recurrent network. ICASSP 2016
31. He K, et al. Deep residual learning for image recognition. CVPR 2016
32. Liu Y. Mutopia: multimodal music understanding. In: ISMIR 2022
33. Devlin J, et al. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805 2018
34. Tsai YHH. Multimodal transformer for unaligned multimodal language sequences. In: ACL 2019
35. Schulman J. Proximal policy optimization algorithms. In: arXiv Preprint arXiv:1707.06347 2017
36. Holstein K. et al. Improving fairness in ai-based education systems. ACM Learning at Scale 2019
37. Bakhtiar A. et al. Trustworthy ai in education: principles for responsible design. Educational Technology Research and Development 2021
38. Cordeiro P, Silva J. Musicai: adaptive music-based tutoring using deep reinforcement learning. IEEE Transactions on Learning Technologies 2021
39. Xu T, Yu J. Reinforcement learning in education: a survey. Computers & Education 2020
40. Moreno R, Mayer RE. Multimedia learning with animated pedagogical agents: effects of voice and appearance. Journal of Educational Psychology 2005
41. Pol J, Volman M, Beishuizen J. Scaffolding the scaffolding metaphor. Educational psychologist 2006
42. Kim K, Lee G, Kim S. A study on the application of blockchain technology in the construction industry. KSCE J Civ Eng. 2020;24(9):2561–71.

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